

# Sustainability aware asset management

## Lecture 13: Financial Regulation and Sustainable Financing Products

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# Objectives of the Lecture

Sustainable investing in equity consists in under- or over-weighting some firms in the portfolio. The reason is that shares cannot be designed to be sustainable independently from the firm itself.

In contrast, it is possible to design investment funds that satisfy some sustainable criteria.

The same applies to bonds, which can specialize in some sustainable themes. In the recent years, several different topics have emerged for bonds:

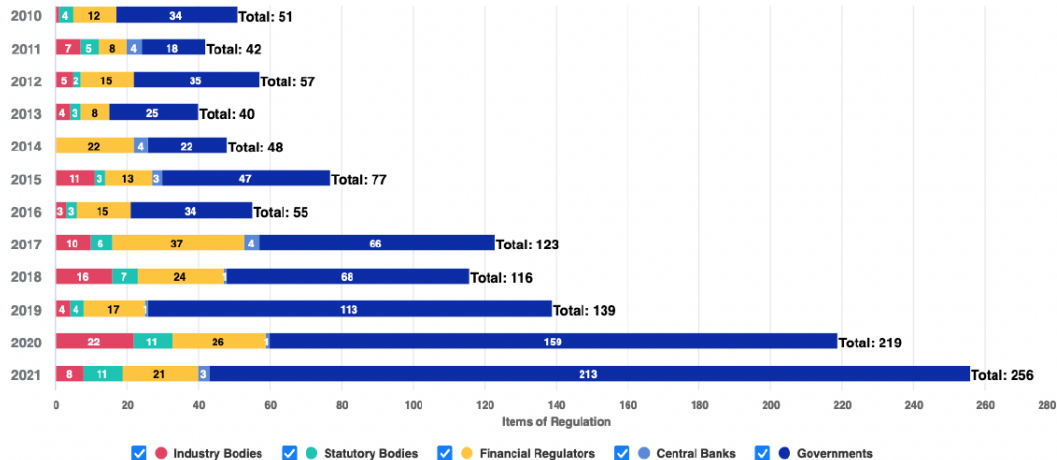
- Green Bonds
- Social Bonds
- Sustainability Bonds
- Sustainability-Linked Bonds
- Transition Bonds

Each category has its own objectives, use of proceeds, and sometimes verification.

# Objectives of the Lecture

- **ESG Regulation**
- SRI Investment Funds
- GSS+ Bonds
- Green Bonds
- Other GSS Bonds
- Sustainability-Linked and Transition Bonds

# ESG Regulation by Regulator



Source: MSCI (2022), "Who will regulate ESG?" (<https://www.msci.com/who-will-regulate-esg>)

The **High-Level Expert Group (HLEG)** on Sustainable Finance was created in October 2016 by the European Commission.

## HLEG 2018 report

- Definition of a taxonomy for sustainable assets
- Inclusion of sustainability and ESG Duties of investors
- Disclosure of ESG metrics
- EU label for green investment funds
- EU standard for green bonds
- Sustainability as part of the mandates of European Supervisory Authorities (ESA)

# ESG Regulation in the European Union

## Systematic and centralized approach toward climate transition and sustainability disclosure

An economic activity must meet 4 conditions to qualify as environmentally sustainable:

- substantially contribute to an environmental objective,
- do no significant harm to other environmental objectives,
- meet minimum social and governance safeguards, and
- comply with technical screening criteria.

New **EU taxonomy** establishes six environmental objectives (July 2020):

- Climate change mitigation
- Climate change adaptation
- The sustainable use and protection of water and marine resources
- The transition to a circular economy
- Pollution prevention and control
- The protection and restoration of biodiversity and ecosystems

# EU ESG Regulation for Asset Managers

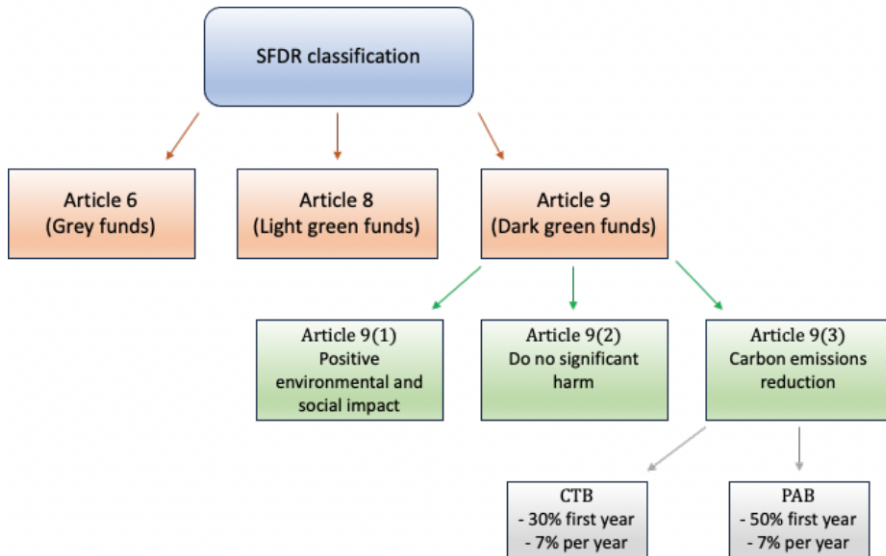
**Sustainable Finance Disclosure Regulation (SFDR)** in Nov. 2019.

Part of the EU's wider Sustainable Finance Framework, SFDR is a set of EU rules which aim to make the sustainability profile of funds more comparable and better understood.

Most visible element: classification of funds and mandates in three categories:

- **Article 6:** covers funds which do not integrate any kind of sustainability into the investment process. These may face marketing difficulties when matched against more sustainable funds.
- **Article 8:** also known as “environmental and socially promoting”, applies “... where a financial product promotes, among other characteristics, environmental or social characteristics, or a combination of those characteristics, provided that the companies in which the investments are made follow good governance practices.”
- **Article 9:** also known as “products targeting sustainable investments”, covers products targeting bespoke sustainable investments and applies “... where a financial product has sustainable investment as its objective and an index has been designated as a reference benchmark.”

# EU ESG Regulation for Asset Managers



**Corporate Sustainability Reporting Directive (CSRD)** in April 2021.

**Objective:** improve the coverage and reliability of sustainability reporting.

- **‘Double materiality’ principle:** Companies must disclose information that is material for the enterprise as well as for its societal stakeholders and/or the environment.
- Example: Companies must disclose the extent to which their activities are compatible with the goal of limiting global warming to 1.5 degrees Celsius.
- Companies must report their sustainability performance using EU-wide disclosure standards.

# ESG Regulation in the United States



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| (GHG emissions in metric tons CO <sub>2</sub> e)    | 2019        | 2020        | 2021        |
|-----------------------------------------------------|-------------|-------------|-------------|
| Scope 1 <sup>1</sup> emissions                      | 105,523,364 | 94,290,023  | 98,749,588  |
| Scope 2 <sup>1</sup> , location-based emissions     | 249,068     | 333,770     | 242,970     |
| Scope 3 <sup>2</sup> emissions                      | -           | 2,724,789   | 2,386,622   |
| Total GHG emissions                                 | 105,772,432 | 97,348,582  | 101,379,180 |
| Total Generation (MWh)                              | 186,428,605 | 172,243,738 | 174,040,471 |
| Emissions Intensity (Scope 1 CO <sub>2</sub> e/MWh) | 0.566       | 0.547       | 0.567       |

$$\text{Carbon intensity} = \frac{87.75 + .0243}{12077} = 8200$$

# ESG Regulation in the United States

Laissez-faire approach with sustainable investing and disclosure being guided by voluntary, private-sector-led processes, protocols, and guidelines.

- Voluntary mechanisms: Corporations are routinely accused of “greenwashing” their sustainability reports by overstating their positive environmental and social impact and downplaying negative ones.
- ESG investors with little assurance, legal or otherwise, that their money has been put to the intended use.

Securities and Exchange Commission (SEC) will probably adopt mandatory ESG disclosure rules, which will be different from EU's approach:

- Regulation will target only publicly listed companies
- SEC will mandate disclosure of a narrow range of outcomes related to climate risk and human capital
- SEC will likely adopt less comprehensive, third-party disclosure standards

# ESG Associations: GSIA

Global Sustainable Investment Alliance (GSIA) (<http://www.gsi-alliance.org>)

## **GSIA members:**

- The European Sustainable Investment Forum (Eurosif)
- Responsible Investment Association Australasia (RIAA)
- Responsible Investment Association Canada (RIA Canada)
- UK Sustainable Investment & Finance Association (UKSIF)
- The Forum for Sustainable & Responsible Investment (US SIF)
- Dutch Association of Investors for Sustainable Development (VBDO)
- Japan Sustainable Investment Forum (JSIF)

Principles for Responsible Investment (PRI) (<https://www.unpri.org>)

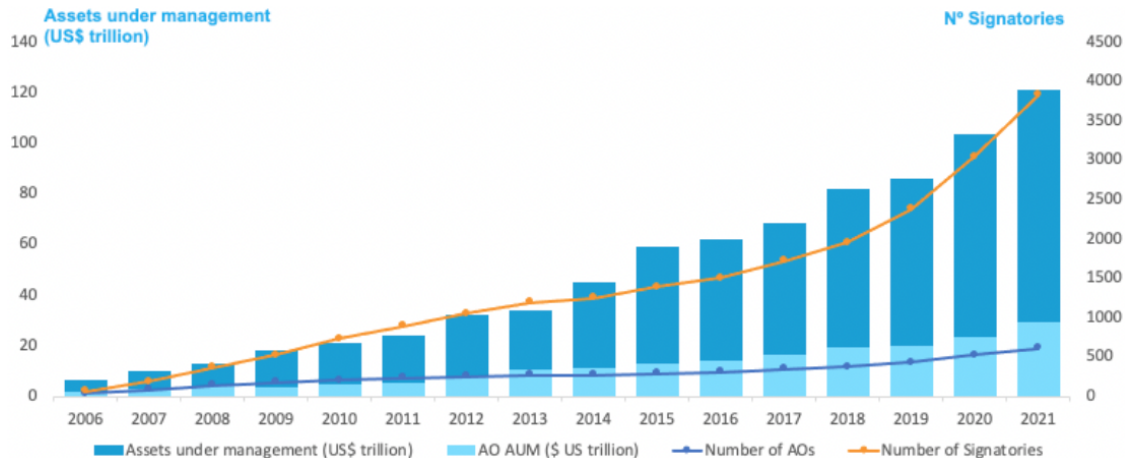
- Early 2005: UN Secretary-General Kofi Annan invited a group of the world's largest institutional investors to join a process to develop the Principles for Responsible Investment
- April 2006: The Principles were launched at the New York Stock Exchange
- 6 ESG principles

# ESG Associations: (UN) PRI

The Principles were developed for investors. Signatories contribute to developing a more sustainable global financial system. They have attracted a global signatory base representing a majority of the world's professionally managed investments.

- **Principle 1:** We will incorporate ESG issues into investment analysis and decision-making processes
- **Principle 2:** We will be active owners and incorporate ESG issues into our ownership policies and practices
- **Principle 3:** We will seek appropriate disclosure on ESG issues by the entities in which we invest
- **Principle 4:** We will promote acceptance and implementation of the Principles within the investment industry
- **Principle 5:** We will work together to enhance our effectiveness in implementing the Principles
- **Principle 6:** We will each report on our activities and progress towards implementing the Principles

# UNPRI: Number of Signatories



Source: UNPRI (2021), <https://www.unpri.org/pri/about-the-pri>. AO: asset owners

# Objectives of the Lecture

- ESG Regulation
- **SRI Investment Funds**
- GSS+ Bonds
- Green Bonds
- Other GSS Bonds
- Sustainability-Linked and Transition Bonds

## Investment vehicles

- Mutual funds
- ETFs
- Mandates & dedicated funds

## Investment strategies

- Thematic strategies (e.g., water, social, wind energy, climate, plastic)
- ESG-tilted strategies (e.g., exclusion, negative screening, best-in-class, enhanced ESG score)
- Climate strategies (e.g., low carbon, 2° alignment, activity exclusions)
- Sustainability-linked securities (e.g., green bonds, social bonds)

Both *active* and *passive* management.

## The big issue for an investor is: How to avoid Greenwashing (& ESG washing)?

### Greenwashing

- Activities by a company or an organization that are intended to make people think that it is concerned about the environment, even if its real business actually harms the environment.
- A common form of greenwash is to publicly claim a commitment to the environment while quietly lobbying to avoid regulation.
- In finance, greenwashing is understood as making misleading claims about environmental practices, performance, or products.

# SRI Investment Funds: Labels

|              | Labels                                                | Governance                                                                                                           | Attribution                                                     | Type of label                                                                                            | Eligibility                                                                               |
|--------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| ESG          | <b>SRI Label</b><br>(France)                          | Standalone stakeholder committee, supported by the Ministry of Finances                                              | Accredited auditors                                             | SRI/ESG investment process                                                                               | UCITS, AIFs (including Real Estate funds)                                                 |
|              | <b>FNG-Siegel</b><br>(Germany, Austria & Switzerland) | Overall steering and review by QNG, a subsidiary of FNG <sup>a</sup> , supervised by an independent expert committee | QNG, based on 3 <sup>rd</sup> -party audits by Univ. of Hamburg | SRI/ESG investment process with ESG and climate exclusions. Point system                                 | Article 8/9 funds distributed in AT/DE/CH/LI                                              |
|              | <b>LuxFLAG ESG</b><br>(Luxembourg)                    | LuxFLAG <sup>b</sup>                                                                                                 | LuxFLAG                                                         | SRI/ESG investment process with climate exclusions                                                       | Article 8/9 funds authorized in an EU country or equivalent authority                     |
|              | <b>Towards Sustainability</b><br>(Belgium)            | Central Labelling Agency <sup>c</sup> (CLA)                                                                          | Verifiers appointed by the CLA                                  | Quality standard combining requirements on the investment process and exclusions                         | All types of funds (Article 8/9 funds if in scope of SFDR)                                |
|              | <b>Umweltzeichen</b><br>(Austria)                     | Austrian Federal Ministry for the Environment                                                                        | Ministry                                                        | SRI/ESG investment process with ESG and climate exclusions. Point system                                 | Investment products, including funds of funds, life insurance products, real estate funds |
| Green labels | <b>Nordic Swan Ecolabel</b><br>(Nordic countries)     | Nordic Ecolabelling Board <sup>d</sup> , on a mandate from Nordic governments                                        | Nordic Swan                                                     | SRI/ESG investment process with ESG and climate exclusions & optional green label criteria. Point system | Article 8/9 funds and investment products that invest in NS labelled funds                |
|              | <b>LuxFLAG Environment</b><br>(Luxembourg)            | LuxFLAG <sup>b</sup>                                                                                                 | LuxFLAG                                                         | Thematic investments and ESG criteria                                                                    | Article 8/9 funds                                                                         |
|              | <b>LuxFLAG Climate Finance</b><br>(Luxembourg)        | LuxFLAG <sup>b</sup>                                                                                                 | LuxFLAG                                                         | Thematic investments and ESG criteria. Climate exclusions                                                | Article 9 funds                                                                           |
|              | <b>Greenfin Label</b><br>(France)                     | Standalone stakeholder committee, chaired by the Ministry for the Ecological Transition                              | Accredited auditors                                             | Thematic investments and ESG criteria. Climate exclusions                                                | Listed UCITS and some non listed AIFs                                                     |

Source: Novethic, “Overview of European sustainable finance labels” (2022)

# SRI Investment Funds: AUM

As of Dec 2020, Green funds in Europe: 130 billion Euro, 340 funds.

|                      | AUM (EUR bln) | Number of funds |
|----------------------|---------------|-----------------|
| Environmental themes | 103           | 232             |
| Low-carbon approach  | 21            | 91              |
| Green bonds          | 18            | 64              |

As of Sept 2021, Sustainable finance labels in Europe: 1'120 billion Euro, 1'600 funds.

|                              | AUM (EUR bln) | Number of funds |
|------------------------------|---------------|-----------------|
| SRI Label                    | 635           | 833             |
| Towards Sustainability Label | 520           | 621             |
| LuxFLAG ESG                  | 154           | 267             |
| FNG Label                    | 81            | 169             |
| Umweltzeichen Ecolabel       | 52            | 201             |
| Nordic Swan Ecolabel         | 35            | 75              |
| Greenfin Label               | 20            | 64              |

Source: Novethic, Market data, sustainable labels Europe, at 30 Sept 2021.

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- Other GSS Bonds
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The Climate Bonds Initiative has defined the GSS+ category of bonds. It includes:

## **Green, social and sustainability (GSS) bonds**

- Green: dedicated environmental benefits (since 2012)
- Social: dedicated social benefits (since 2020)
- Sustainability: green and social benefits combined into one instrument (since 2020)

## **Sustainability-linked bonds (SLB)**

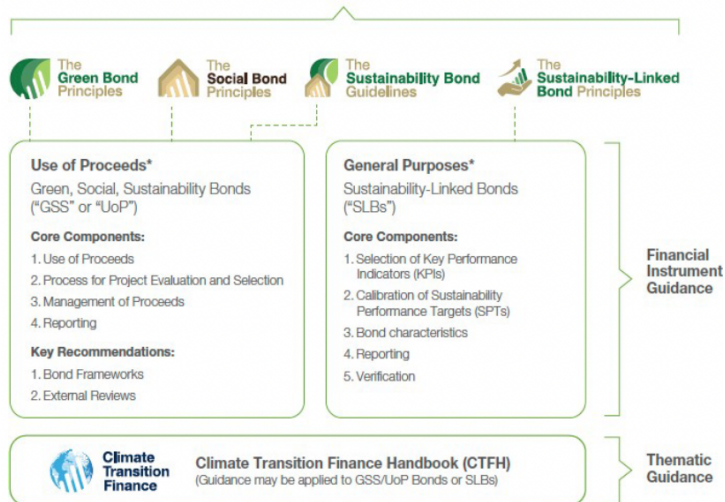
- Changes in coupon linked to performance against entity level sustainability performance targets (since 2021).

## **Transition bonds**

- Use of Proceeds supporting transition at activity or entity level (since 2021).

Source: CBI (2019), <https://www.climatebonds.net/>.

## The Principles

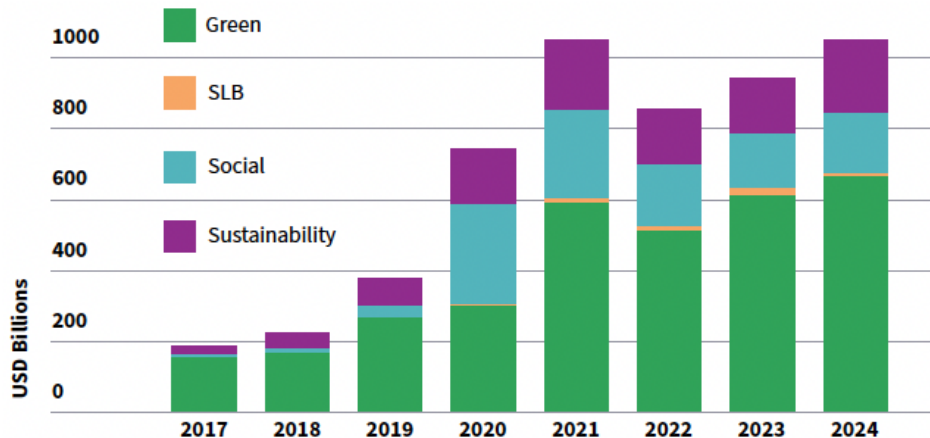


Source: ICMA (2021), Sustainability Bond Guidelines

| At the end of 2022            | Green | Social | Sustainability | SLB   | Transition |
|-------------------------------|-------|--------|----------------|-------|------------|
| <b>Cumulative size</b>        |       |        |                |       |            |
| Total size of market (\$ bln) | 2,200 | 653.6  | 682.0          | 204.2 | 12.5       |
| Number of issuers             | 2,457 | 772    | 507            | 336   | 39         |
| Number of countries           | 85    | 49     | 57             | 50    | 12         |
| Number of currencies          | 49    | 42     | 41             | 21    | 7          |
| <b>2022</b>                   |       |        |                |       |            |
| New issuance (\$ bln)         | 487.1 | 130.3  | 161.3          | 76.4  | 3.5        |
| Number of issuers             | 741   | 153    | 206            | 142   | 25         |
| Number of countries           | 51    | 27     | 36             | 35    | 3          |
| Number of currencies          | 33    | 24     | 28             | 16    | 3          |

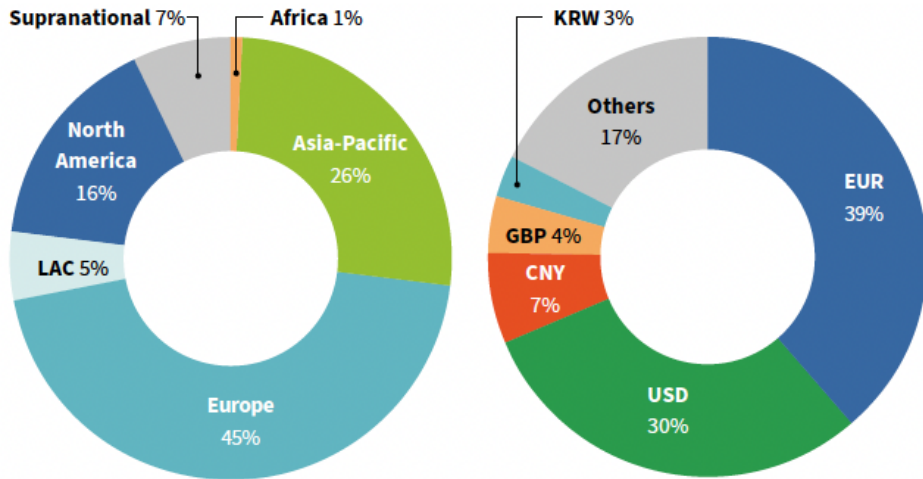
Source: Climate Bonds Initiative (2023), *Sustainable debt, Global state of the market 2022*.

# GSS+ Market – Volumes



Source: Climate Bonds Initiative (2023), "Sustainable debt: Global state of the market 2022"

# GSS+ Market – Countries



Source: Climate Bonds Initiative (2023), "Sustainable debt: Global state of the market 2022"

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# Green Bonds – Definition

Green bonds (or green loans/green debt instruments) are debt instruments where the proceeds will be exclusively applied to finance or re-finance, in part or in full, new and/or existing eligible green projects, and which is aligned with the four core components of the Green Bond Principles (GBP) or the Green Loan Principles.

Source: CBI (2019), <https://www.climatebonds.net/>.

Green bonds are “regular” bonds aiming at funding projects with positive environmental and/or climate benefits.

# Green Bonds Principles

## Green Bonds Principles (GBP).

The 4 core components of the GBP are:

- Use of proceeds
  - Pollution prevention and control
  - Biodiversity conservation
  - Climate change adaptation
- Process for project evaluation and selection
- Management of proceeds
- Reporting

Source: <https://www.icmagroup.org/sustainable-finance/the-principles-guidelines-and-handbooks>.

## Use of proceeds (Component 1):

- Renewable energy
- Energy efficiency
- Pollution prevention (e.g., GHG control, soil remediation, waste recycling)
- Sustainable management of living natural resources (e.g., sustainable agriculture, sustainable forestry, restoration of natural landscapes)
- Terrestrial and aquatic biodiversity conservation (e.g., protection of coastal, marine and watershed environments)
- Clean transportation
- Sustainable water management
- Climate change adaptation
- Eco-efficient products
- Green buildings

## Process for project evaluation and selection (Component 2)

The issuer of a green bond should clearly communicate:

- the environmental sustainability objectives
- the eligible projects
- the related eligibility criteria

## Management of proceeds (Component 3)

It includes:

- the tracking of the “balance sheet” and the allocation of funds
- an external review (not mandatory but highly recommended)

**Reporting (Component 4)** must be based on the following pillars:

- Transparency
- Description of the projects, allocated amounts and expected impacts
- Qualitative performance indicators
- Quantitative performance measures (e.g., energy capacity, electricity generation, GHG emissions reduced/avoided, number of people provided with access to clean power, decrease in water use, reduction in the number of cars required)

# Green Bonds: Market Players

## Green bond issuers

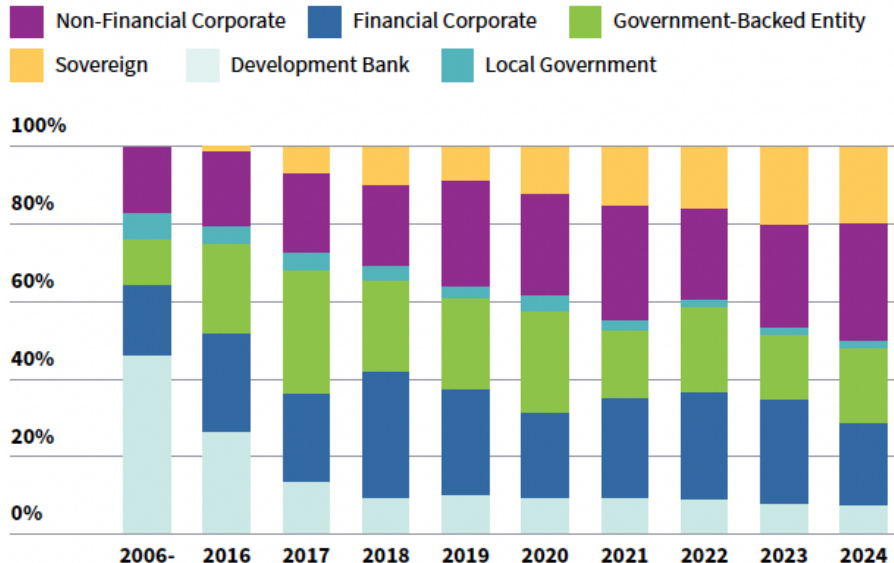
- Sovereigns (agencies, municipals, governments)
- Multilateral development banks (MDB)
- Energy and utility companies
- Banks
- Other corporates

## Green bond investors

- Pension funds
- Sovereign wealth funds
- Insurance companies
- Asset managers
- Retail investors (e.g., employee savings plans)

**Strong imbalance between supply and demand.**

# Green Bonds: The Market – Issuers



# Investing in Green Bonds – Active management

Actively managed green bond funds (largest AUM):

| Name                                            | Currency<br>Base | Launch<br>date | AUM (\$mln)<br>12-2020 |
|-------------------------------------------------|------------------|----------------|------------------------|
| Eurizon Absolute Green Bonds Z Cap              | EUR              | 10.01.18       | 2455.4                 |
| NN (L) Green Bond I Cap EUR                     | EUR              | 01.03.16       | 1667.3                 |
| Amundi Planet - Emerging Green One - Senior USD | USD              | 28.02.18       | 1561.8                 |
| BNP Paribas Green Bd I Cap (formerly Parvest)   | EUR              | 07.09.17       | 1318.7                 |
| Allianz Green Bond W EUR                        | EUR              | 17.11.15       | 1002.6                 |
| Amundi Rspnb Investing Imp Gr Bds I C           | EUR              | 27.09.16       | 919.1                  |

Source:

<https://www.environmental-finance.com/content/analysis/actively-managed-green-bond-funds-table.html>.

# Investing in Green Bonds – Passive management

## Green bond indices

- Bloomberg Barclays MSCI Global Green Bond Index
- S&P Dow Jones Green Bond Index
- Solactive Green Bond Index
- ChinaBond China Climate-Aligned Bond Index
- Bank of America Merrill Lynch Green Index

## Green bond ETFs and index funds (end of 2022)

| ETF name                             | Currency base | Index                                                 | Launch date | Size at launch | AUM (\$mln) 12-2022 |
|--------------------------------------|---------------|-------------------------------------------------------|-------------|----------------|---------------------|
| Lyxor Green Bond DR UCITS ETF        | EUR           | Solactive Green Bond                                  | 02.17       | 5m             | 587.2               |
| iShares Global Green Bond ETF        | USD           | Bloomberg Barclays MSCI Global Green Bond             | 11.18       | 25m            | 73.5                |
| Franklin Liberty Euro Green Bond ETF | EUR           | Bloomberg Barclays MSCI Euro Green Bond               | 04.19       | 10m            | 291.1               |
| Van Eck Vectors Green Bond ETF       | USD           | S&P Green Bond Select                                 | 03.17       | 5m             | 184.9               |
| Lyxor Green Bond ESG Screened        | EUR           | Solactive Green ESG Bond EUR USD IG                   | 10.19       | 4m             | 154.8               |
| UC MSCI European Green Bond ETF      | EUR           | Bloomberg Barclays MSCI European GB Issuer Capped EUR | 11.18       | 20m            | 211.6               |

Source: Climate Bonds Initiative (2022), “Green bond pricing in the primary market: H2 2022”.

# Green Bond Premium

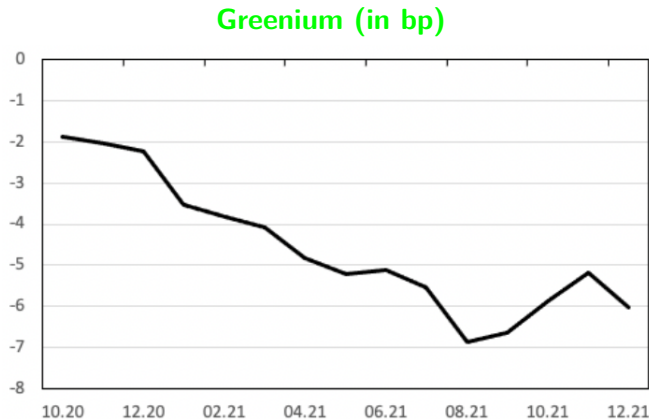
The green bond premium (or **greenium**) is the difference in pricing between green bonds and regular bonds.

- From the **issuer's point of view**, a green bond issuance is more expensive than a conventional issuance due to the need for external review, regular reporting, and impact assessments.
- From the **investor's point of view**, there is no fundamental difference between a green bond and a conventional bond, meaning that one should consider a negative green bond premium as a market anomaly. Some investors are willing to pay a premium for green bonds, rewarding companies or governments that want to clean up their act by giving them lower borrowing costs.

The greenium seems to be at least partially due to the oversubscription of green bonds (demand pressure), through the inflows in green ETFs.

# Green Bond Premium: German Twin Bonds

Difference (in bp) between the yield of the 10-year German Federal Government bond issued in Sept 2020 (maturity 15.08.2030, 0% coupon, green bond) and the 10-year German Federal Government bond issued in June 2020 (maturity 15.08.2030, 0% coupon, regular bond)



Source: Pastor, Stambaugh, Taylor (2021), "Dissecting green returns"

# Objectives of the Lecture

- ESG Regulation
- SRI Investment Funds
- GSS+ Bonds
- Green Bonds
- **Other GSS Bonds**
- Sustainability-Linked and Transition Bonds

# Social Bonds – Definition

Social Bonds are any type of bond instrument where the proceeds will be exclusively applied to finance or re-finance in part or in full new and/or existing eligible Social Projects and which are aligned with the four core components of the Social Bonds Principles (SBP).

## Social Bonds Principles (SBP)

The 4 core components of the SBP are:

- Use of proceeds
- Process for project evaluation and selection
- Management of proceeds
- Reporting

Source: ICMA (2020), <https://www.icmagroup.org/sustainable-finance>.

# Social Bonds Principles

## Eligible social project categories (Component 1)

- Affordable basic infrastructure (clean drinking water, sanitation, clean energy, ...)
- Access to essential services (health, education)
- Affordable housing (sustainable cities)
- Employment generation (pandemic crisis)
- Food security and sustainable food systems (nutritious and sufficient food, resilient agriculture)
- Socioeconomic advancement and empowerment (income inequality, gender

## Target populations (Component 1)

- Living below the poverty line
- Excluded and/or marginalized populations/communities
- People with disabilities
- Migrants and /or displaced persons
- Undereducated
- Unemployed
- Women and/or sexual and gender minorities
- Aging populations and vulnerable youth

# Sustainability Bonds – Definition

Sustainability Bonds are bonds where the proceeds will be exclusively applied to finance or re-finance a combination of both green and social projects.

Sustainability Bonds are aligned with the four core components of both the GBP and SBP with the former being especially relevant to underlying Green Projects and the latter to underlying Social Projects.

# Objectives of the Lecture

- ESG Regulation
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- **Sustainability-Linked and Transition Bonds**

# Sustainability-Linked and Transition Bonds

## Definition: Sustainability-Linked Bonds

Any type of bond for which the financial and/or structural characteristics can vary depending on whether the issuer achieves predefined Sustainability/ESG objectives. Issuers are thereby committing explicitly to future improvements in sustainability outcome(s) within a predefined timeline. SLBs are a forward-looking performance-based instrument.

## Definition: Transition Bonds

Allow high emitters to fund their shift towards cleaner, more sustainable operations and strategies on the way to net zero. When thoughtfully constructed, these debt instruments can be pivotal in supporting a global, economy-wide transition to the Paris Agreement targets.

Source: Climate Bonds Initiative (2021), Sustainable debt market Summary Q3 2021.

# Sustainability-Linked Bond Principles (SLBP)

The 5 core components of the SLBP are:

- Selection of Key Performance Indicators (KPIs)
- Calibration of Sustainability Performance Targets (SPTs)
- Bond characteristics
- Reporting
- Verification

Source: <https://www.icmagroup.org/sustainable-finance/the-principles-guidelines-and-handbooks>.

# Sustainability-Linked Bonds Principles

## Selection of KPIs (Component 1).

The KPIs should be:

- relevant, core and material to the issuer's overall business, and of high strategic significance to the issuer's current and/or future operations
- measurable or quantifiable on a consistent methodological basis
- externally verifiable
- able to be benchmarked, i.e., as much as possible using an external reference or definitions to facilitate the assessment of the SPT's level of ambition

## Calibration of Sustainability Performance Targets (SPTs) (Component 2).

The SPTs should be ambitious:

- represent a material improvement in the respective KPIs and be beyond a "Business as Usual" trajectory
- where possible be compared to a benchmark or an external reference
- be consistent with the issuers' overall strategic sustainability / ESG strategy
- be determined on a predefined timeline, set before (or concurrently with) the issuance of the bond

# Sustainability-Linked Bonds Principles

**Bond characteristics (Component 3)** should vary depending on whether the selected KPI(s) reach (or not) the predefined SPT(s), i.e. the SLB will need to include a financial and/or structural impact involving trigger event(s).

**Reporting (Component 4)** must include:

- up-to-date information on the performance of the selected KPI(s)
- a verification assurance report relative to the SPT outlining the performance against the SPTs and the related impact (and its timing) on the bond's characteristics
- any information enabling investors to monitor the level of ambition of the SPTs (e.g., any update in the issuers sustainability strategy)

**Verification (Component 5)** should be made by a qualified external reviewer (auditor or environmental consultant), at least once a year, and made publicly available.

# Sustainability-Linked Bonds Principles

## **Chile: The world's first sovereign SLB issuer – coupon step-up linked to the NDC and to gender equality**

In February 2022, Chile issued the world's first sovereign SLB. Its stated purpose was to “leverage ambitious timelines to achieve strong sustainable outcomes that are relevant, core and material to Chile and the Chilean people” (Public Debt Office of the Ministry of Finance of Chile, 2023<sup>[17]</sup>). Chile built on its previous GSS issuances, with the aim to inspire other sovereigns and corporates to do the same (Public Debt Office of the Ministry of Finance of Chile, 2023<sup>[17]</sup>). Its SLB, which matures in 2042, raised USD 2 billion and achieved very high and diversified investor demand of USD 8.1 billion (GFL - Green Finance for Latin America and the Caribbean, 2022<sup>[18]</sup>).

The SLB has two KPIs that, alongside the associated SPTs, are in line with the environmental objectives set in Chile's updated NDC. The bond has an associated coupon step-up, which “raises the coupon by 12.5 basis points per KPI, from 4.34% per annum at issuance to 4.59% if both KPIs underperform the SPT at the penalty event date” (Kulenkampff and Pipan, 2023<sup>[19]</sup>).

In 2023, Chile added a third KPI to its SLB Framework, focused on gender equality and women's empowerment. In June of that year, Chile issued a USD 3 billion multi-currency package of SLBs composed of a USD 1.15 billion 12-year SLB, a USD 1.1 billion 30-year SLB, and a EUR 750 million 11-year SLB (Table 1.2). Not only was this the first sovereign SLB to incorporate social targets, but it was also the first sovereign SLB in euros (Bruni and Lewis, 2023<sup>[14]</sup>). This bond also has an associated coupon step-up.

Source: Paul Horrocks (2024), “Sustainability-linked bonds: How to make them work in developing countries, and how donors can help”, OECD working paper

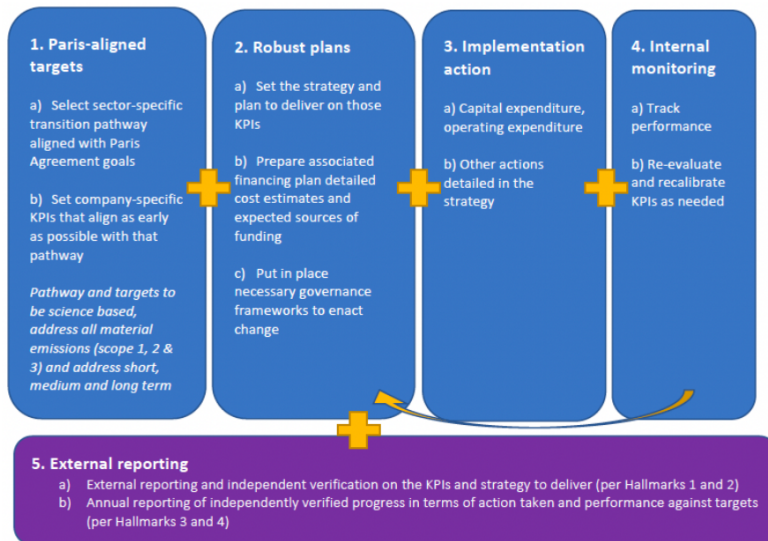
# Sustainability-Linked Bonds Principles

**Table 1.2. Characteristics of the 2023 Chile Sustainability-linked Bond Framework**

| KPI                                                                                                                                                  | Associated SPT                                                                                                  | Observation date | Structural and/or financial characteristic(s)                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Greenhouse gas emissions (GHG) per year, measured in metric tonnes of CO <sub>2</sub> equivalent (MtCO <sub>2</sub> e)                            | Achieve GHG emissions of 95 MtCO <sub>2</sub> e by 2030                                                         | 31 December 2030 | Chile will link the bond's financial characteristics to the achievement of the SPTs, such as a coupon step-up or a premium payment in case an SPT, including an intermediate SPT, is not met at the target observation date. In the event that more than one SPT is not met, the premium paid will be cumulative |
|                                                                                                                                                      | Achieve a maximum of 1 100 MtCO <sub>2</sub> e between 2020 and 2030                                            | 31 December 2030 |                                                                                                                                                                                                                                                                                                                  |
| 2. Non-conventional renewable energy (NCRE) generation as the percentage generated in the national electric system, measured in megawatt hours (MWh) | Achieve 50% electric generation derived from NCRE sources by 2028                                               | 31 December 2028 |                                                                                                                                                                                                                                                                                                                  |
|                                                                                                                                                      | Achieve 60% electric generation derived from NCRE sources by 2032                                               | 31 December 2032 |                                                                                                                                                                                                                                                                                                                  |
| 3. Percentage of women in board member positions at companies that report to Chile's Financial Market Commission (CMF)                               | Achieve at least 40% of women representation in boards of directors at companies that report to the CMF by 2031 | 31 December 2031 |                                                                                                                                                                                                                                                                                                                  |

Source: Public Debt Office of the Ministry of Finance of Chile, (2023<sub>[17]</sub>), *Chile's Sustainability Linked Bond Framework: June 2023*, <https://www.hacienda.cl/english/work-areas/international-finance/public-debt-office/esg-bonds/sustainability-linked-bonds/chile-s-slb-framework-june-2023-version>; Sustainalytics (2023<sub>[20]</sub>), *Second-Party Opinion: Chile's Sustainability-linked Bond Framework*, <https://www.hacienda.cl/english/work-areas/international-finance/public-debt-office/esg-bonds/sustainability-linked-bonds/second-party-opinion-sustainalytics-2023->

# Transition Bonds



Source: Climate Bonds Initiative (2021) Sustainable debt market Summary Q3 2021

End of Lecture 13